Netting Hub Test Plan

1. Introduction

This document defines the test plan for the Netting Hub project. The solution contains three core modules:

1. FX Netting
2. TD Netting
3. FOF (Fund of Funds / or any third module applicable to the project)

The goal of this test plan is to validate that each module works correctly, integrates with the overall platform, and supports business-critical netting operations with accurate calculations, reliable processing, and proper exception handling.

2. Project Objective

The Netting Hub is expected to:

- collect trade and exposure data from various sources
- perform netting calculations by module
- support exception tracking and approvals
- ensure data integrity and auditability
- provide reporting and monitoring for operational teams
- support reconciliation with downstream and upstream systems

3. Scope

In Scope

- FX Netting processes and calculations
- TD Netting processes and calculations
- FOF processing and calculations
- Data validation and transformation
- Business rule validations
- Reconciliation and exception management
- Workflow approvals and status tracking
- Reporting and dashboards
- Security and role-based access
- Integration with external systems

Out of Scope

- Full end-to-end production deployment
- Performance testing for extreme volumes beyond approved benchmarks
- Non-functional testing beyond defined acceptance criteria

4. Modules Overview

4.1 FX Netting

This module handles foreign exchange netting activities, including:

- trade matching
- currency pair validation
- netting schedule processing
- settlement instructions
- margin or exposure checks

4.2 TD Netting

This module handles trade or treasury derivative netting activities, including:

- counterparty netting
- exposure aggregation
- settlement group creation
- threshold and tolerance checks
- exception and dispute management

4.3 FOF

This module manages the Fund of Funds or other third functional area, including:

- fund allocation logic
- portfolio grouping
- netting of transactions across fund structures
- exception resolution
- reporting by fund or pool

5. Test Objectives

- Verify that all business rules are implemented correctly.
- Validate that each module produces correct netting results.
- Ensure integration between modules and shared data services works correctly.
- Confirm system behavior under valid and invalid inputs.
- Validate robustness of exception handling.
- Ensure usability and operational readiness for business users.

6. Test Strategy

The project will use a combination of the following test levels:

6.1 Functional Testing

Validate that business functions behave as expected.

6.2 Integration Testing

Validate communication and data flow between:

- modules within Netting Hub
- downstream systems
- upstream source feeds
- reporting and settlement services

6.3 Regression Testing

Retest impacted functions after each defect fix or change.

6.4 User Acceptance Testing (UAT)

Business users validate end-to-end business scenarios before sign-off.

6.5 Performance Testing

Validate expected response times and transaction processing under normal and peak load.

6.6 Security Testing

Validate role-based access, authorization, and audit trail.

7. Test Environment

- Application environment: development / QA / UAT / production-like staging

- Database: test instance with masked and synthetic data
- Integration endpoints: mock or real dependent systems
- Reporting tools: dashboard/reporting environment
- User roles: business analyst, operations, risk, support, admin

8. Entry Criteria

- Requirements reviewed and approved
- Test data prepared
- Test environment available and stable
- Users and roles created
- Defect triage process active
- Build deployed to test environment

9. Exit Criteria

- All critical and high-severity defects resolved or formally accepted by stakeholders
- All planned test cases executed
- Regression suite passed
- UAT sign-off obtained
- No unresolved showstopper issues

10. Assumptions and Risks

Assumptions

- Business requirements are finalized and stable
- Data feeds are available in the test environment
- Users will participate in UAT and defect review sessions
- Access to all modules is configured before testing starts

Risks

- Incomplete or changing requirements
- Data quality issues from upstream systems
- Delayed environment setup
- Integration dependency failures
- Insufficient test data coverage for edge cases

11. Test Data Requirements

Test data should cover:

- valid and invalid trade records
- multiple counterparties
- different currencies and settlement dates
- duplicate and partial records
- zero/negative balances and tolerance conditions
- exceptions, disputes, and approvals
- edge cases around netting threshold and limit logic

12. Test Scenarios

12.1 FX Netting Scenarios

- Create valid trade set for netting
- Process cross-currency netting
- Validate rounding and rate conversion
- Handle unmatched trade pairs
- Validate settlement date matching

Process failed or rejected trade entries
Verify exception generation and resolution

12.2 TD Netting Scenarios

Aggregate position across multiple trades
Apply netting threshold and tolerance rules
Validate counterparty grouping logic
Process multi-leg settlement instructions
Review exception queue and approval workflows
Recalculate adjusted net exposure after modification

12.3 FOF Scenarios

Allocate transactions across funds or portfolios
Validate group-level aggregation
Handle missing or invalid fund mappings
Process netting across multiple fund structures
Verify proper reporting by fund and portfolio
Validate approval and exception handling for unusual allocations

13. Detailed Test Cases

13.1 FX Netting Test Cases

ID	Test Case	Description	Expected Result
FX-01	Create valid netting request	Submit valid FX trades for matching	Trades are matched and net exposure calculated
FX-02	Invalid currency pair	Submit unsupported currency pair	Validation error raised and transaction rejected
FX-03	Duplicate trade record	Upload duplicate trade	Duplicate record flagged and handled per rule
FX-04	Settlement date mismatch	Trades with different settlement dates	System detects mismatch and routes to exception
FX-05	Threshold validation	Trades below threshold	Not netted or routed based on configured rule
FX-06	Rate conversion	Convert trades using configured rate	Correct final net amounts generated
FX-07	Exception resolution	Fix invalid trade after exception	Trade resubmitted and processed successfully

13.2 TD Netting Test Cases

ID	Test Case	Description	Expected Result
TD-01	Create valid TD netting batch	Submit valid trade batch	Netting run completes successfully
TD-02	Counterparty grouping	Multiple trades under same counterparty	Netting grouped correctly
TD-03	Tolerance check	Trade amount within tolerance	Netting accepted as configured
TD-04	Threshold breach	Exceeds threshold	Exception raised for review
TD-05	Partial cancellation	Cancel some trades in netting batch	Remaining positions re-netted correctly
TD-06	Approval workflow	Submit for approval	Status moves to pending/approved/rejected correctly
TD-07	Reconciliation	Compare netted output with source data	Output matches approved reconciliation rules

13.3 FOF Test Cases

ID	Test Case	Description	Expected Result
FOF-01	Valid fund allocation	Submit trades with valid fund mapping	Allocation and netting processed successfully
FOF-02	Missing fund mapping	Submit fund with no mapping	System creates validation exception
FOF-03	Multi-fund aggregation	Transactions across multiple funds	Totals aggregated by fund and portfolio
FOF-04	Duplicate allocation	Duplicate fund allocation records	Duplicate flagged and not double-counted
FOF-05	Approval path	Raise FOF exception for review	Workflow routes to designated approver
FOF-06	Reporting output	Generate fund-level report	Report reflects accurate netted balances
FOF-07	Exception closure	Resolve mismatch or data issue	Output recalculated and approved

14. Integration Test Cases

FX module sends valid settlement data to downstream settlement service
TD module updates net position in central exposure repository

- FOF module shares allocation data with reporting data mart
- Error messages propagate correctly across modules
- Audit logs are generated for all status changes

15. Non-Functional Requirements to Validate

- Response time for netting job execution
- Throughput under expected peak load
- Data consistency across modules
- Audit trail completeness
- Role-based access restrictions
- Error logging and traceability

16. Defect Management

- Defects logged with severity and priority
- Critical defects must be fixed before UAT sign-off
- Defects tracked through triage, resolution, and retest
- Business stakeholders informed of high-risk issues

17. Roles and Responsibilities

- Business Analyst: requirements validation
- Test Lead: overall planning and coordination
- QA Engineer: test execution and defect logging
- Developers: bug fixing and code support
- Product/Operations Owner: UAT validation and sign-off
- Security/Admin: access and role setup

18. Test Execution Schedule

This is a sample indicative schedule:

- Requirement review and test planning: Week 1
- Test data preparation: Week 1-2
- Functional testing: Week 2-3
- Integration testing: Week 3-4
- UAT: Week 4-5
- Defect retest and sign-off: Week 5

19. Acceptance Criteria

The project is considered ready for sign-off when:

- all critical and high-severity defects are closed or accepted
- all planned test cases are executed
- business processes are validated in UAT
- reports and reconciliations match expected results
- operational teams confirm readiness

20. Sign-off

This test plan is subject to approval by:

- Project Manager
- Functional Lead
- Technical Lead
- Business Owner
- QA Lead

21. Summary

The Netting Hub test plan covers the expected business and technical validation for FX Netting, TD Netting, and FOF. The focus is on verifying accurate netting logic, integration quality, operational readiness, and user confidence before production release.